MZUZU UNIVERSITY

DEPARTMENT OF MATHEMATICS AND STATISTICS

MATH 1101 – PRECALCULUS

FINAL EXAM

Date: February 11, 2019 Time Allowed: 3 hours (14:00 – 17:00 HRS)

INSTRUCTIONS

- Write your registration number, programme and lecturer's name in the spaces provided below
- Check that the paper has **12** questions
- Attempt **all** questions
- Write your answer in the space provided under each question
- Show all your working
- Calculators are not allowed

Registration Number:

Programme:

Lecturer's Name:

Question:	1	2	3	4	5	6	7	8	9	10	11	12	Tota
Points:	6	6	8	8	11	5	6	7	10	15	7	11	100
Score:													

1. Reduce the following fraction to its lowest form. $\boxed{6}$

$$\frac{\left(a+\frac{1}{b}\right)^a\left(a-\frac{1}{b}\right)^b}{\left(b+\frac{1}{a}\right)^a\left(b-\frac{1}{a}\right)^b}.$$

2. Find the value of k such that the following equation has exactly one root. $\boxed{6}$

$$3x^2+(\sqrt{2k})x+6=0.$$

5. Express the domain of the following functions in interval notation.

(a) $f(x) = \dfrac{4}{2 - \sqrt{x+3}}$. $\boxed{6}$

(b) $h(x) = \log_{10} \dfrac{2x+3}{x-5}$. $\boxed{5}$

6. Let $f(x) = 2x^3 + 1$. Find $f^{-1}(x)$. $\boxed{5}$

3. Solve the following inequality and put your answer in interval notation. $\boxed{8}$

$$\left|\frac{x+1}{2} - \frac{x-1}{3}\right| < 1.$$

4. Solve the equation $\sqrt{x+2} + \sqrt{x-2} = \sqrt{4x-2}$. $\boxed{8}$

7. Let $f(x) = -2x+1$ and $g(x) = ax+b$. Find a and b so that the equation $(f \circ g)(x) = x$ holds for all values of x. $\boxed{6}$

8. Consider the function $y = 2x^2+3x-2$. Find the vertex, minimum or maximum value and intercept(s) of its graph. $\boxed{7}$

9. (a) Simplify $2^{\log_2 5} - 3\log_5 \sqrt[3]{5}$. $\boxed{4}$

(b) Solve for x: $\log_2(\log_3 x) = -1$. $\boxed{6}$

10. Evaluate each expression.

(a) $\csc \frac{4\pi}{3}$. $\boxed{5}$

(b) $\tan\left(-\frac{\pi}{3}\right)$. $\boxed{5}$

(c) $\cos \frac{13\pi}{3}$. $\boxed{5}$

11. Verify the identity $\boxed{7}$

$$\frac{\sin\theta}{1+\cos\theta} + \frac{1+\cos\theta}{\sin\theta} = 2\csc\theta.$$

12. Sketch the graph the following function on the interval $-\pi \leq \theta < 2\pi$. Specifying the amplitude, period, x-intercept, and intervals on which the graph is increasing. $\boxed{11}$

$$y = 2\sin\left(x - \frac{\pi}{2}\right).$$

END OF QUESTION PAPER